Camphene to Camphor

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Chem 2081 001

Introduction

The purpose of this experiment was to synthesize camphor in a multistep synthesis beginning with camphene. The initial step was the acid-catalyzed electrophilic addition of camphene to give isobornyl acetate, followed by the conversion to borneol.

Materials and Methods

First, 0.3 g of the borneol product with 0.3 mL glacial acetic acid was added into a long-neck, round-bottom flask along with a boiling chip. Next, a reflux with a distillation column was assembled, and the apparatus was placed in a water bath with a temperature ranging from 40° C - 50° C. After the reagents were dissolved, 6 mL of aqueous sodium hypochloride solution was added slowly and the solution was agitated. The reaction was allowed to continue for 15 minutes after the last addition of sodium hypochloride. The mixture was cooled, transferred to a centrifuge tube, and the camphor was extracted with methylene chloride. The combined methylene layers were neutralized via 2 mL of saturated sodium bicarbonate. Next, the organic solution was washed with 2 mL of 5% sodium bisulfite and 2 mL of water. During each extraction, the bottom layer is able to be rinsed over the top due to the organic layer being more dense than the other layer. A possible source of error can come from extracting the product after separation of layers. Doing an extraction step incorrectly can corrupt the purity of the final product. The solution was then dried over anhydrous sodium sulfate. Lastly, the DCM was evaporated over the water bath and weighed.

Calculations

$Borneal\ \frac{0.3\ g}{154.35\ mol}\ =\ 0.002$

$Camphor\ 0.002\ \times\ 152.23\ =\ 0.296g$

$\%\ yield\ =\ \frac{0.02}{0.296} = 67.57\%$

Results

Mass of Reaction Product	0.296 g
Percent Yield	67.57%

Table 1. Mass of Reaction Product and Percent Yield

Discussion

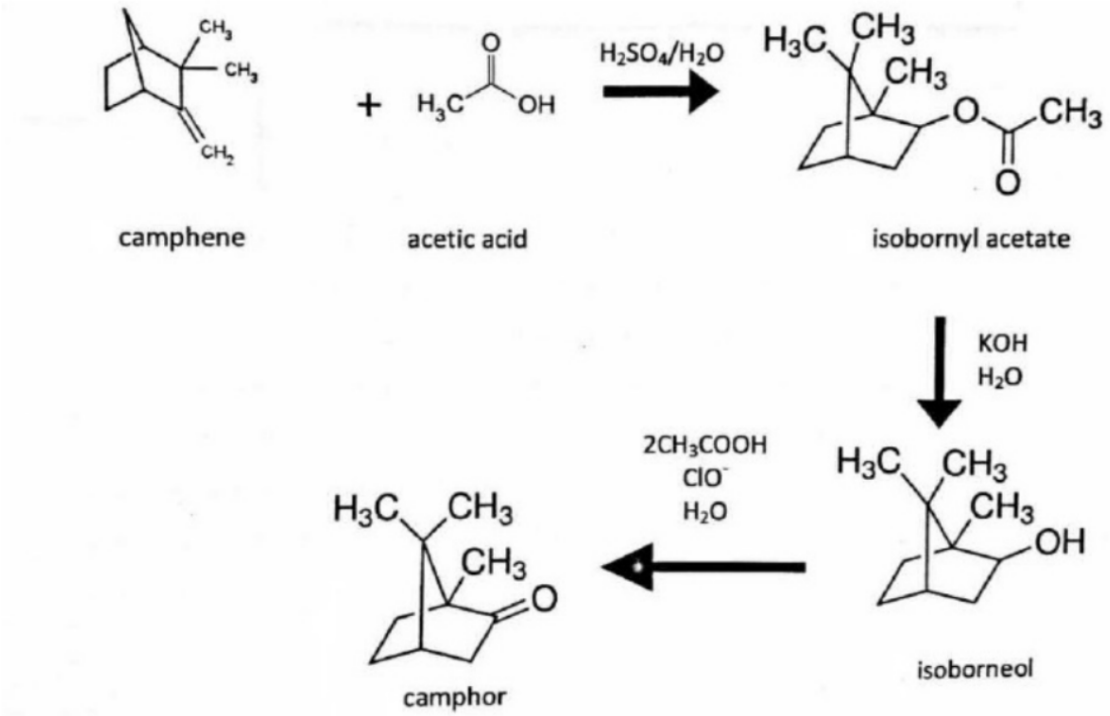
First, 0.3 g of the racemic borneol product with 0.3 mL glacial acetic acid was added into a long-neck, round-bottom flask along with a boiling chip. A racemic mixture is a solution with a 50:50 enantiomer ratio. A reflux with a distillation column was assembled alongside a water bath ranging from 40°C - 50°C. The reagents dissolved within a few minutes, so no acetone was added to the solution. Once the reagents were dissolved, 6 mL of aqueous sodium chloride solution (bleach) was added slowly over the span of 15 minutes. After each addition, the solution was agitated to ensure the chemicals mixed thoroughly. The reaction was allowed to continue an additional 15 minutes after the last addition of bleach. The mixture was then allowed to cool and the solution was transferred to a centrifuge tube for better extraction using DCM. The DCM weighs 1.33 g/mL making it the bottom layer of the solution. The layers were then neutralized by saturated sodium bicarbonate. Carbon dioxide bubbles were observed

after the addition. Next, the solution was washed with 2 mL of 5% sodium bisulfate and 2 mL of water and then dried with anhydrous sodium sulfate. The DCM was then evaporated over the water bath ad weighed

References

Keller L, Joseph K, and Miguel S. 2023 Fall. CHEM 2081 Organic Chemistry 2 Lab Manual [lab manual]. Auburn (Al): Auburn University.

Mechanism


camphene
acetic acid
H_2SO_4/H_2O
isobornyl acetate
KOH
H_2O
isoborneol
$2CH_3COOH$
ClO^-
H_2O
camphor